Visitor readings at Hertenstein: protocol

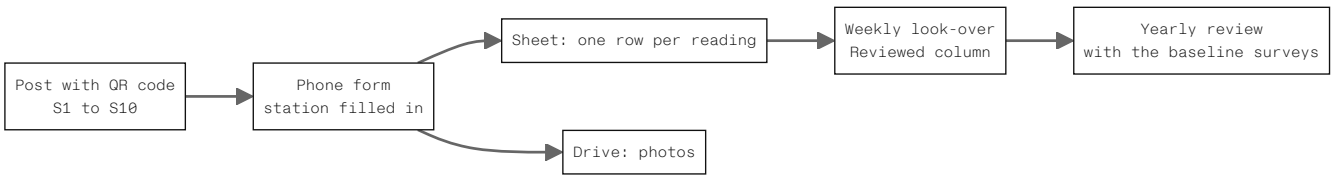
First draft, 13 October 2026. Companion to hertenstein-ecology-master-plan-2026-10-13.md.

What it is

Anyone on the site can take a short reading of the land or the lake with their phone. Each reading takes between one and fifteen minutes, needs no training, and adds a row to one shared record. Over years, readings at the same places in the same way show how the site is changing between professional surveys.

How it works

1. **Ten stations.** Each is a wooden post with a number (S1 to S10) and a QR code. The code opens the form with the station already filled in.
2. **Pick a reading.** The form shows the readings that suit that station first.
3. **Do it, then send.** Numbers, a few choices, optional photos. No login.
4. **Where it goes.** One row per reading in a Google Sheet, photos in a Drive folder, both in the stella.earth Drive. Nothing is published automatically.
5. **Once a week** someone from the garden group or Stella looks down the new rows, marks anything doubtful in the Reviewed column, and moves good photos of plants and animals to iNaturalist if the observer agreed.



The stations (provisional)

Exact positions are set on a site walk, ideally with whoever does the vegetation plots so stations sit next to permanent plots. Positions get logged once with a phone GPS and kept in the Stations tab of the Sheet.

Station	Habitat	Readings that suit it
S1	Oak park, under the oldest oak	Birds, first-of-year, anything noticed
S2	Best meadow (least fertilised)	Flowers, pollinators, birds
S3	Ordinary hay meadow (for comparison with S2)	Flowers, pollinators, soil
S4	Kitchen garden	Soil, pollinators, weather
S5	Woodland edge	Birds, invasives, anything noticed
S6	Hedge or orchard	Birds, pollinators, first-of-year
S7	Shore or landing	Lake, birds, invasives

Station	Habitat	Readings that suit it
S8	Wettest ground or pond, if there is one	Anything noticed, first-of-year
S9	Open spot away from lights	Night sky, bats, weather
S10	Roof terrace or courtyard near the house	Weather, birds, first-of-year

The readings

Each reading has a fixed method so that two people at the same station on different days are measuring the same thing. Methods are printed in the form itself in one or two lines.

1. Five-minute bird count

Stand at the post. Set a five-minute timer. Write down every bird you see or hear and how many, the highest number seen at once for each kind. Unknown birds are fine: "small brown bird, 2" still counts. If you have the Merlin app, its sound ID can help, but write only what you are fairly sure of.

Records: species and counts, total, heard or seen, confidence.

2. Ten-minute pollinator count

Pick one patch of flowers about the size of an A3 sheet (30 × 40 cm), of a single kind if possible. Note the flower. Set a ten-minute timer and count every insect that lands on a flower in the patch, by group: bumblebees, honeybees, other bees, hoverflies, other flies, butterflies and moths, wasps, beetles, other. Only on dry days above 13 °C with little wind. Adapted from the UK Pollinator Monitoring Scheme's Flower-Insect Timed count.

Records: flower, counts per group, sun, wind, number of flowers in the patch.

3. Flowers in bloom

Lay a one-metre square (a four-metre loop of string with knots at each metre, kept at the post) on the meadow beside the post. Count how many different kinds of flower are in bloom inside it. Grasses do not count. Name them if you can; a photo of each helps. Measure the height of the grass in centimetres at the middle.

Records: number of kinds, names, grass height, photos.

4. Lake

At the landing (S7). Water temperature with the thermometer kept at the post, held 20 cm under the surface for one minute. Clarity with the Secchi disc if there is one (lower it on its cord until it disappears, note the depth in metres). Algae: none, some, lots. Mussels: lift three stones near the edge and note if any small striped mussels are attached (zebra or quagga mussel). Water level from the gauge if one is fitted.

Records: temperature, clarity, algae, mussels, level.

5. Soil

At S3 or S4, in the marked soil square only. Dig one spade-deep block about 20 × 20 cm, break it up on a sheet, count the earthworms, put everything back. Then the water test: push the tin ring kept at the post 5 cm into the ground nearby, pour in one litre of water, time how long until it has all sunk in. Note how the soil feels: crumbly, clumpy, hard.

Records: worms, seconds to soak, texture, moisture.

6. Night

At S9, at least an hour after sunset, no moon if possible, after five minutes for your eyes to adjust. Count the stars you can see inside the square of the constellation shown on the form for that season (or choose the Globe at Night chart that matches). Count the artificial lights you can see from the post. Note any bats.

Records: stars, chart match, lights, bats, cloud.

7. First of the year

The first time this year you see: the kites back (the Sisters recorded them on 19 March), a swallow or swift, the first cuckoo heard, oak leaves opening, the first hay cut, first frost, first snow on the Rigi. One sighting per event per year is enough; the earliest one counts.

Records: event, date, place.

8. Weather

Air temperature from the thermometer in the shade, rain from the gauge since last reading (empty it after), wind, cloud.

Records: temperature, rain, wind, cloud.

9. Invasive plant or pest

You see one of: cherry laurel, narrow-leaved ragwort (yellow daisy-like flowers, narrow leaves), Armenian blackberry, annual fleabane, black locust, Japanese knotweed, Himalayan balsam, goldenrod, or box-tree moth caterpillars on box. Take a photo, say how much (one plant, a patch under 1 m², 1 to 10 m², more). **Please don't pull plants yourself**: some are toxic and some spread when cut. The garden group will deal with them.

Records: species, extent, photo, location.

10. Anything else

Something unusual: a dead animal, an orchid, a nest, a fallen branch on a path, erosion, rubbish in the water. Photo and a line of text.

Data and privacy

- Name and email are optional. Readings without a name are fine.
- Location comes from the station, or from the phone if the visitor taps "use my location" and the browser allows it. Photo files keep any location the phone stored.
- Photos of people are not wanted. The form says so.
- **Rare species stay private.** Orchids, nests and roosts are never published with their location. If a reading is shared (iNaturalist, a report), sensitive locations are blurred to at least 1 km.
- Visitors choose whether a photo can be shared with credit, shared without credit, or kept only for the site record.

Quality

- Every reading records the observer's experience (first time, done it before, expert) and their confidence. Analysis can weight or filter on these.
- Same method, same place, every time. The form shows the method beside the fields.
- The Reviewed and Review note columns in the Sheet are for the weekly look-over. Doubtful readings get marked, never deleted.
- Once a year, compare visitor readings with the professional survey at the same stations to see what the readings catch and miss, and adjust the protocol.

Kit at the stations

Station	Kit
All	Post with number and QR code; small waterproof box
S2, S3	String square (4 m loop, knotted at each metre); ruler
S3 or S4	Marked soil square; tin ring (15 cm across, 10 cm deep); one-litre jug
S7	Thermometer on a cord; Secchi disc on a marked cord; lake level gauge (needs the operator's and possibly the canton's agreement)
S9	Laminated star charts for each season
S4 or S10	Shaded thermometer; rain gauge

The string square, tin ring and jug can sit in the box at the house instead if posts would be vandalised or untidy.

What it needs before it goes live

1. Sylvan's yes to deploy the form (setup steps in readings-form/SETUP.md).
2. A site walk to place the ten stations, with the owner side's agreement on posts and kit.
3. QR codes printed for each station link.
4. One named person for the weekly look-over (garden group or Stella).
5. A decision on whether to also run an iNaturalist project for the site, which would make photo records public with blurred locations for sensitive species.